Multimodal contrastive learning and weak supervision

Two modalities, one embedding space

CHANGE WHAT COUNTS AS A POSITIVE PAIR

Single modality (SimCLR)

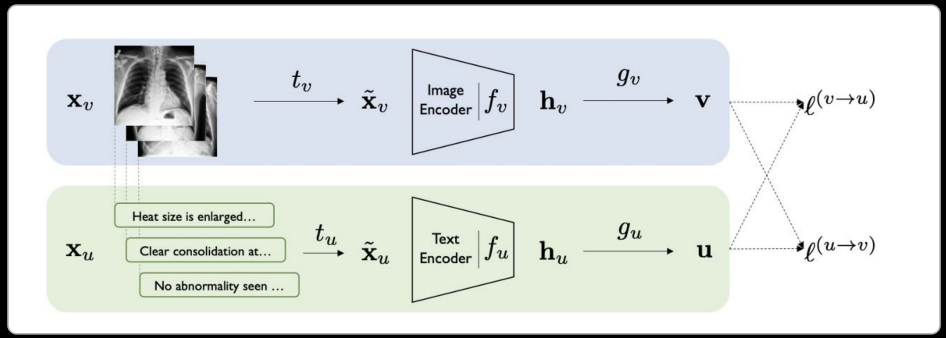
A positive pair is **two augmentations of one image**, encoded by the **same** encoder. The model learns what survives a transformation.

Two modalities (ConVIRT, CLIP)

A positive pair is **an image and its own report**, encoded by **two different** encoders. The model learns what the two modalities agree about.

The loss does not change at all. Only the definition of "these two belong together" changes, and with it the supervision source: from a transformation we chose, to a **pairing the hospital produced for free**.

ConVIRT: contrastive pre-training on images and reports



You met ConVIRT in Lecture 1 as an example of **coordination**; here is how it works. **217k** image and text pairs from MIMIC-CXR. A **ResNet-50** encodes the image, a **BERT** text encoder initialized from ClinicalBERT encodes the report, each with its own projection head. The loss runs in both directions, image to text and text to image, so neither modality is merely a target.

Zhang et al., *Contrastive Learning of Medical Visual Representations from Paired Images and Text*, MLHC. BERT is a transformer text encoder; we use it here as a black box.

Nobody annotated anything

Where the supervision came from

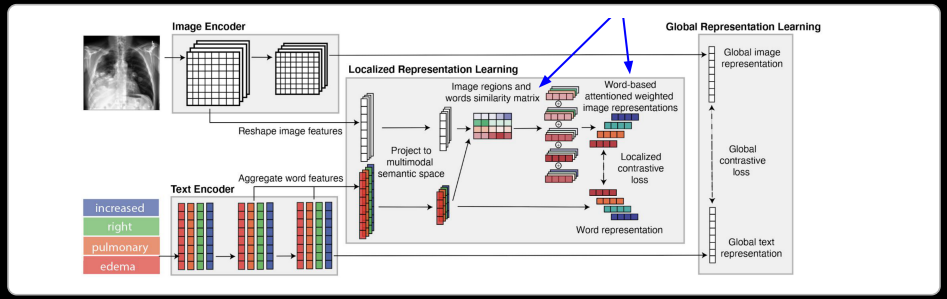
The radiologist wrote the report **for patient care**, years before anyone thought of training a model on it. The image and the text were already filed together.

What that changes

Scale stops being an annotation budget and becomes a **data access** question. Every study in the archive is a training pair.

This is the central argument of the lecture. The most valuable labelled dataset in a hospital is the one nobody labelled on purpose. It is also, in the next section, a source of **noisy labels** rather than just pairings.

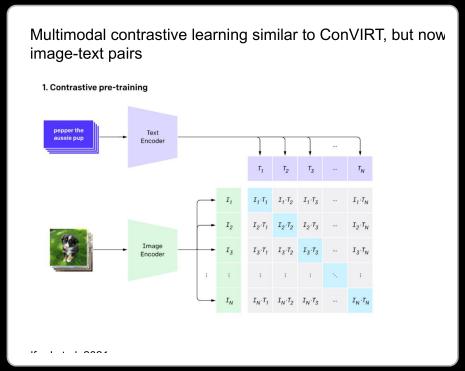
GLoRIA: REPORTS ARE LONG, IMAGES ARE LARGE



Matching a whole report to a whole image is coarse: “small left pleural effusion” describes one corner of one lung, not the study. GLoRIA keeps the global contrastive loss and adds a **localized** one, between individual **words** and attention-weighted **image regions**, learned jointly.

Huang et al., *GLoRIA*, ICCV

CLIP: THE SAME IDEA AT WEB SCALE



What is different

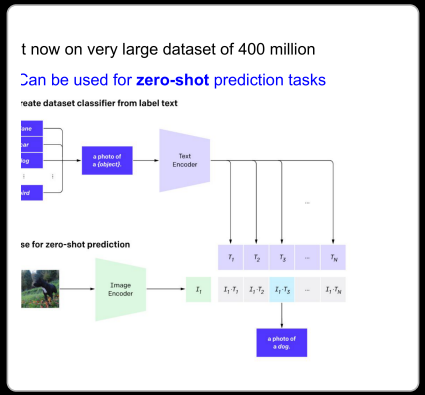
400 million image and text pairs scraped from the web, transformer encoders on both sides trained from scratch. Every image in the batch is scored against every caption, and the diagonal is the set of true pairs.

What is the same

The objective. ConVIRT and CLIP are the same loss with different data and different budgets.

Radford et al., *Learning Transferable Visual Models From Natural Language Supervision*, ICML

Zero-shot prediction, and what it means in radiology



Build a classifier out of sentences

Write one sentence per class, encode them with the text encoder, and pick the class whose sentence is closest to the image. **No task-specific training and no labels at all.**

The honest caveat

Impressive on natural images. On clinical findings it is uneven, sensitive to how the sentence is phrased, and nowhere near a supervised model on a task you actually care about.

Radford et al., ICML

Labels you do not trust

Weak supervision

The paradigm

Obtain labels for supervised training from **noisy or imprecise** sources that are much easier to access than expert annotation, then train on those labels while accounting for their unreliability.

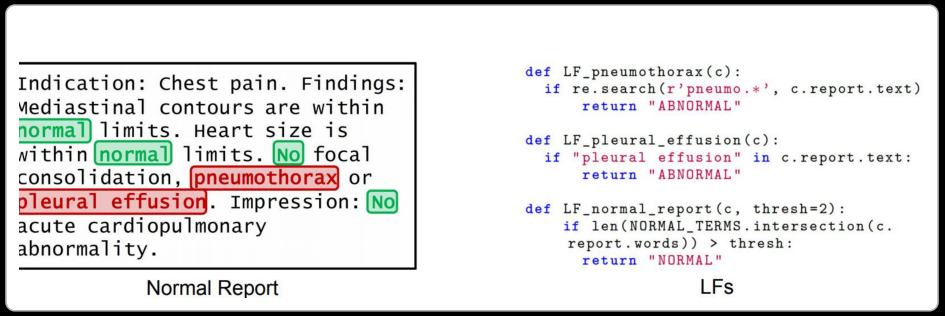
Where the noise comes from

Heuristics, rules, ontologies, crowd workers, billing codes, and **another modality**: the report that describes the image.

How it relates to self-supervision

Complementary to self-supervision, not an alternative. Self-supervision gives you a good encoder; weak supervision gives you *labels* to fine-tune it on.

LABELING FUNCTIONS OVER REPORTS



Each labeling function is a few lines of code over the free text: a regular expression for `pneumo.*`, a search for "pleural effusion", a count of normal terms. Each returns **ABNORMAL**, **NORMAL**, or abstains. Individually they are crude; a clinician would reject any one of them.

after Khandwala et al.; Dunmon et al., *Cross-Modal Data Programming*, Patterns

From noisy votes to a usable label

Naive: majority vote

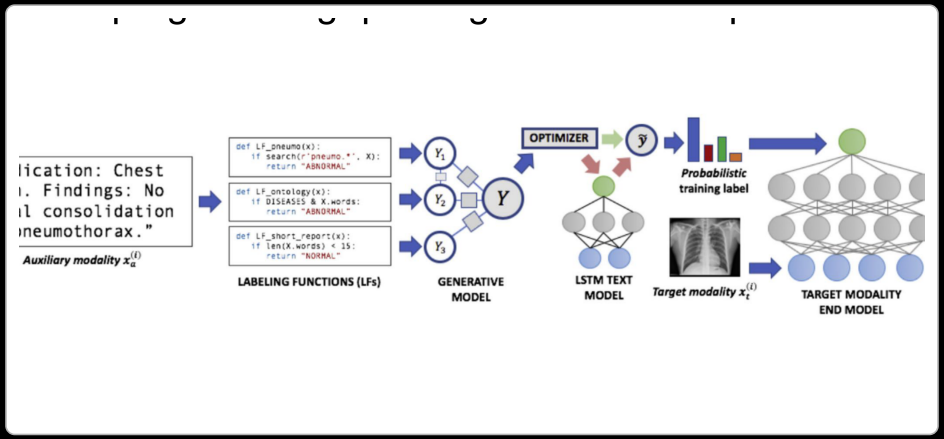
Run every labeling function, take the majority. Simple, and wrong whenever the functions are correlated: three rules that all grep for the same word are one rule counted three times.

Better: learn the aggregation

Fit a **generative model** over how the functions agree, disagree and correlate, with the true label as a latent variable. It infers each function's accuracy **without ever seeing a gold label**.

The output is not a hard label but a **probabilistic** one, and the downstream model is trained against those probabilities, so an uncertain example contributes proportionally less.

CROSS-MODAL DATA PROGRAMMING, END TO END



Write rules over the **auxiliary modality** (the report), aggregate them into probabilistic labels, and train the end model on the **target modality** (the image). At deployment the report is gone: the image model stands alone, trained on labels nobody wrote by hand.

Dunmon et al., *Cross-Modal Data Programming Enables Rapid Medical Machine Learning*, Patterns

Three things to remember

1. Fusion is a choice of **depth**: early combines data, joint combines features, late combines predictions. Real models mix them, and the right depth follows from how far apart the modalities are.

2. **Self-supervision** turns an unlabelled archive into an encoder. The contrastive loss that does it is the cross-entropy you already know, with “correct class” redefined as “the other view”.

3. The report you already have is **both** a free positive pair for contrastive pre-training and a free noisy label for weak supervision. The most useful annotations in a hospital were never made for you.

All three of this lecture arc’s ideas answer the question Module 2 opened: what do you do when labels are scarce.

References

Multimodal models

- Huang et al. *Fusion of medical imaging and EHR using deep learning*. npj Digital Medicine, 2020.
- Havaei et al. *Brain Tumor Segmentation with DNNs*. Medical Image Analysis, 2016.
- Wu et al. *DNNs Improve Radiologists' Performance in Breast Cancer Screening*. IEEE TMI, 2019.
- Wang et al. *TieNet*. CVPR 2018.
- Rajkomar et al. *Scalable and accurate deep learning with EHRs*. npj Digital Medicine, 2018.

Self- and weak supervision

- Chen et al. *SimCLR*. ICML 2020.
- Zhang et al. *ConVIRT*. 2020.
- Huang et al. *GLoRIA*. ICCV 2021.
- Radford et al. *CLIP*. ICML 2021.
- Dunmon et al. *Cross-Modal Data Programming Enables Rapid Medical Machine Learning*. Patterns, 2020.